

Smartphone Ownership, Age of Smartphone Acquisition, and Health Outcomes in Early Adolescence

Ran Barzilay, MD, PhD,^{1,2,3} Samuel D. Pimentel, PhD,⁴ Kate T. Tran, MPH,^{1,2} Elina Visoki, MSc,^{1,2} David Pagliaccio, PhD,⁵ Randy P. Auerbach, PhD⁵

abstract

OBJECTIVES: Given concerns regarding health implications of adolescent smartphone use, we tested associations of smartphone ownership and age of smartphone acquisition with depression, obesity, and insufficient sleep in early adolescence. We hypothesized that smartphone ownership, especially at a younger age, would be associated with worse health outcomes.

METHODS: The sample included 10 588 participants from the Adolescent Brain Cognitive Development Study. Mixed-effects logistic regression models tested associations of smartphone ownership and age of first smartphone acquisition, reported by caregivers, with depression, obesity, and insufficient sleep at age 12 years. Among participants who did not own smartphones at age 12 years, we tested associations of recent acquisition of smartphones with outcomes in the subsequent year. Models were adjusted for demographic and socioeconomic variables, ownership of other devices, pubertal development, and parental monitoring.

RESULTS: At age 12 years, compared with not owning a smartphone ($n = 3849$), smartphone ownership ($n = 6739$) was associated with higher risk for depression (odds ratio [OR] 1.31, 95% CI: 1.05–1.63), obesity (OR 1.40, 95% CI: 1.20–1.63), and insufficient sleep (OR 1.62, 95% CI: 1.46–1.79). Younger age of smartphone acquisition was associated with obesity and insufficient sleep (for each earlier year of acquisition, OR 1.09, 95% CI: 1.02–1.16, and OR 1.08, 95% CI: 1.02–1.12, respectively). At age 13 years, among 3486 youth who did not own a smartphone at age 12 years, those who had acquired a smartphone in the past year ($n = 1546$) had greater odds of reporting clinical-level psychopathology (OR 1.57, 95% CI: 1.12–2.20) and insufficient sleep (OR 1.50, 95% CI: 1.26–1.77) compared with those who had not ($n = 1940$) after controlling for baseline mental health and sleep. Results were consistent across several sensitivity analyses.

CONCLUSIONS: Smartphone ownership was associated with depression, obesity, and insufficient sleep in early adolescence. Findings provide critical and timely insights that should inform caregivers regarding adolescent smartphone use and, ideally, the development of public policy that protects youth.



¹Department of Child and Adolescent Psychiatry and Behavioral Sciences, Children's Hospital of Philadelphia, Philadelphia, Pennsylvania; ²Lifespan Brain Institute of Children's Hospital of Pennsylvania and Penn Medicine, Philadelphia, Pennsylvania; ³Department of Psychiatry, Perelman School of Medicine, University of Pennsylvania, Philadelphia, Pennsylvania; ⁴Department of Statistics, University of California Berkeley, Berkeley, California; and ⁵Department of Psychiatry, Columbia University, Vagelos College of Physicians and Surgeons, New York, New York

Address correspondence to: Ran Barzilay, MD, PhD, 10th Floor, Gates Pavilion, Hospital of the University of Pennsylvania, 34th and Spruce St, Philadelphia, PA 19104.
ran.barzilay@pennmedicine.upenn.edu

Dr Barzilay contributed to study conceptualization, formulation of research questions and study design, supervision, and writing of the first draft of the manuscript. Dr Pimentel contributed to study conceptualization, study design, statistical methodology, supervision, and provision of critical revision of the manuscript. Ms Tran and Ms Visoki contributed to data curation, data analysis, data visualization, and critical revision of the manuscript. Dr Pagliaccio contributed to conceptualization of study, interpretation of findings, and critical revision of the draft of the manuscript. (Continued)

WHAT'S KNOWN ON THIS SUBJECT: Most American youths acquire a smartphone during early adolescence. Previous research has linked general screen use around this age, but not specifically smartphone use, to impairments in mental health, body weight, and sleep patterns.

WHAT THIS STUDY ADDS: Among US youths of early adolescent age, smartphone ownership and earlier age of smartphone acquisition is associated with higher rates of depression, obesity, and insufficient sleep. These findings highlight the need to consider health risks when teens gain smartphone access.

To cite: Barzilay R, Pimentel SD, Tran KT, et al. Smartphone Ownership, Age of Smartphone Acquisition, and Health Outcomes in Early Adolescence. *Pediatrics*. 2026;157(1):e2025072941

INTRODUCTION

Smartphone ownership is a *rite of passage* for youths, and, in the United States, approximately one-half of children will have their own smartphone by age 11 years.¹ Smartphones provide ready and immediate access to schoolwork, social networks, internet browsing, gaming, and social media. Given the potential for negative exposures (eg, bullying, explicit content) coupled with the fact that many 8- to 12-year-old individuals are spending approximately 4 to 6 hours per day using screens, inclusive of smartphones,² and the potential adverse health associations of screen time,^{3,4} there is growing concern regarding how smartphone use affects mental and physical health among young people. Accordingly, the American Academy of Pediatrics (AAP) and the American Academy of Child and Adolescent Psychiatry (AACAP) have established guidelines to support families in instituting healthier boundaries, delineating recommended screen time limits, preferred parental controls, and optimal parental modeling practices.⁵ To our knowledge, however, there are no guidelines regarding the appropriate age at which youths should first own a smartphone.^{6,7}

The decision to provide a child with their own smartphone is often initiated as a matter of necessity, allowing communication with caregivers and addressing safety concerns. Although the intended use for smartphones may be circumscribed, there is a natural drift wherein youth begin to use their smartphones for other activities (eg, socialization, education).⁶ Most prior research has focused on the consequences of excessive, addictive, or problematic smartphone use,^{8–12} reflecting a preoccupation and relative loss of control, reported among 3% to 35% of youths.¹³ Problematic smartphone use is associated with a wide range of negative outcomes, including adverse mental health consequences (eg, depression), increased sedentary behaviors, and diminished sleep.^{9,14,15} However, it is not evident whether simply owning a smartphone, accounting for the age of ownership, in childhood and early adolescence potentiates risks for negative health consequences.

As smartphone ownership among children becomes more ubiquitous, there is a critical need for data to inform families regarding the decision of when to provide their child with access to smartphones, as well as to guide public health policy. Thus, this study aimed to examine how acquiring a smartphone relates to health outcomes in early adolescence. Accordingly, we used data from the Adolescent Brain Cognitive Development (ABCD) Study, which tracks US youths annually from late childhood. We hypothesized that smartphone ownership and earlier acquisition of smartphones would be associated with depression, obesity, and insufficient sleep.

METHODS

Participants

ABCD Study participants were enrolled across 21 sites throughout the United States and were assessed from late childhood into adolescence.¹⁶ Therefore, the baseline assessment was completed at a mean age of approximately 10 years, with follow-up assessments each subsequent year (eg, 1-year follow-up at a mean age of approximately 11 years, 2-year follow-up at a mean age of approximately 12 years, 3-year follow-up at a mean age of approximately 13 years). The ABCD Study conducts in-person structured clinical computerized assessments, whereas body mass index (BMI) is measured by taking height and weight of youth study participants. ABCD data release 5.1, which included 5 assessments conducted between 2016 and 2022 (baseline to 4-year-follow-up, age range of 9–16 years), was used. Smartphone ownership was first assessed in the 2-year follow-up assessment; therefore, we used data from this assessment and the subsequent assessments. Sample description is summarized in Supplemental Table S1. Note that data on less than one-half of the sample were available for the 4-year follow-up assessment. These data were used only in supplemental sensitivity analyses. The ABCD Study protocol was approved by the University of California San Diego institutional review board (IRB), and analyses were exempted from review by the Children's Hospital of Philadelphia IRB. We adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines.

Smartphone Ownership

Smartphone ownership was assessed annually starting at the 2-year follow-up assessment based on caregiver response to the question “Does your child have their own cellphone?” and to the follow-up question regarding device type. Participants with an iPhone or Android were considered as owning a smartphone. Age of receiving a smartphone was identified based on caregiver response to the question “How old was your child when he/she got their own cellphone?” First-time smartphone ownership in the past year was considered for youths whose caregiver responded “no” regarding cellphone ownership at the 2-year follow-up but later responded “yes” at the 3-year follow-up.

Health-Related Outcomes

To enhance clinical interpretability and effect size interpretation, we used binary outcomes in our main analyses. Mental health outcomes included lifetime depression diagnosis based on the “Kiddie Schedule for Affective Disorders and Schizophrenia Child Psychiatric Diagnostic Interview” (KSADS)¹⁷ at the 2-year and 4-year follow-up assessments

and on clinical-threshold level (T-score ≥ 65)¹⁸ of psychopathology at the 3-year follow-up (which did not include KSADS assessment) based on the self-reported Brief Problem Monitor (BPM)¹⁹ scale, which assesses internalizing, externalizing, and attention symptoms.¹⁸ Obesity was determined by BMI (kilograms per meters squared, based on weight/height measurements during study assessment) greater than the 95th percentile.²⁰ Insufficient sleep was defined as youth-reported sleep duration of less than 9 hours per night.²¹ Supplemental Material eMethods provide further details on outcome measures.

Statistical Analyses

We conducted 3 sets of analyses to test our hypotheses. Analysis 1 tested associations of smartphone ownership with outcomes at age 12 years (2-year follow-up). Analysis 2 examined how the age at which participants first acquired a smartphone (in years; continuous measure) was associated with health outcomes at age 12 years, focusing specifically on the subgroup that already owned a smartphone by that age. Analysis 3 investigated the associations between first acquiring a smartphone in the past year and outcomes at age 13 years. Analysis 3 included only participants without smartphones at age 12 years, creating a cohort that compared those who subsequently acquired smartphones vs those who remained without smartphones by age 13 years, ensuring that both groups shared a common preexposure period. This allows preexposure outcome measures, assessed at age 12 years (psychopathology, obesity, insufficient sleep), to be adjusted for as covariates, providing a higher quality of causal evidence.

Mixed-effects logistic regression models (“mixed models”) used smartphone ownership (Analysis 1), age of first smartphone acquisition (Analysis 2), or past-year smartphone acquisition (Analysis 3) as independent variables, with depression, obesity, and insufficient sleep as outcome variables. Analysis 3 used clinical-threshold psychopathology as the outcome instead of depression because the KSADS was not used at the 3-year follow-up assessment. Mixed models accounted for the hierarchical nature of the data using nested random effects for family (when siblings were enrolled) and research site. To correct for the testing of 3 outcomes, we applied a *P* value threshold of 0.017 (0.05/3). For each outcome, we estimated a base model that included age and sex assigned at birth and an adjusted model that covaried for confounders, including household income, parental education, participants’ race and ethnicity (caregiver report), pubertal status (pre-, early, mid-, late, and post-pubertal)²², and parental monitoring (average of 5 youth-reported items,²³ each rated from 1, never, to 5, always),²³ which is associated with the studied outcomes.^{24–26} To enhance specificity of results to smartphone ownership, this model adjusted for ownership of other devices (eg, tablet, laptop, iPod, smartwatch).

Covariates are described in the Supplemental Material eMethods. Note that the rationale to include race and ethnicity as covariates²⁷ was intended to address potential confounding owing to differences in smartphone ownership (exposure variable) and health outcomes among racial and ethnic groups in the ABCD Study.^{28,29} We explored sex differences by adding the interaction term (product) of the exposure and sex to the adjusted model. Models used listwise deletion of missing data. Data missingness is provided in Tables 1 and 2. Because no analysis included multiple time points with smartphone ownership, adjustments for dropout and intention-to-treat analyses were not required. Comparison of participants who were included vs those who were excluded because of missing exposure data (<10%) is provided in Supplemental Table S2.

For each model, we report the odds ratio (OR), exponentiated regression coefficient, 95% CI value, and two-sided *P* value. In Analysis 2, coefficients were transformed to reflect a positive relationship between younger age and adverse outcome (depression, obesity, insufficient sleep) by using the inverse of the OR and 95% CI values. We used R (version 4.5.0, R Project for Statistical Computing) package lme4 for statistical models and ggplot package for data visualization.

Sensitivity Analyses

First, we re-estimated models excluding participants with problematic smartphone use. Because there is no gold-standard tool to quantify problematic smartphone use,³⁰ we used the Mobile Phone Involvement Questionnaire,³¹ which probes problematic smartphone use patterns in the ABCD Study³² (detailed in the Supplemental Material eMethods).

Because our analysis of observational data depends on an assumption of no unobserved confounding between exposed vs unexposed conditions, we conducted E-value sensitivity analysis³³ to quantify the degree of unobserved confounding required to explain our results in the absence of a treatment effect (detailed in the Supplemental Material eMethods).

We re-estimated models with continuous outcome measures (overall psychopathology, BMI, hours of sleep) using linear mixed models.

Finally, we re-estimated mixed models of Analyses 1 and 2 using longitudinal data from the 2-year, 3-year, and 4-year follow-up assessments.

RESULTS

Smartphone Ownership

Among youths who reported owning a cellphone, nearly all (6739 of 7010; 96.1%) owned a smartphone (iPhone or Android; Table 1). Smartphone ownership rose steadily across the assessments: 63.6% (6739 of 10 588) at the

TABLE 1. Comparison of Youths Who Own a Smartphone With Those Who Do Not by the 2-Year Follow-Up Assessment Wave					
	Overall Sample	Does Not Own a Smartphone	Owens a Smartphone	Uncorrected P Value	Missing Data, %
n	10 588	3849	6739	NA	NA
Smartphone type, iPhone	NA	NA	4313	NA	NA
Smartphone type, Android	NA	NA	2426	NA	NA
Demographic characteristics					
Age at 2-y follow-up, mean (SD)	12.02 (0.67)	11.83 (0.63)	12.13 (0.66)	<.001	0.01
Late/post-pubertal (2-y follow-up), n (%)	2122 (20.04)	480 (12.47)	1642 (24.37)	<.001	5.37
Female sex assigned at birth, n (%)	5028 (47.49)	1623 (42.17)	3405 (50.53)	<.001	0.03
White race, n (%)	7985 (75.42)	3161 (82.13)	4824 (71.58)	<.001	0.17
Black race, n (%)	2153 (20.33)	492 (12.78)	1661 (24.65)	<.001	0.17
Asian, n (%)	668 (6.31)	272 (7.07)	396 (5.88)	.02	0.17
American Indian and Alaska Native race, n (%)	365 (3.45)	97 (2.52)	268 (3.98)	<.001	0.17
Mixed race, n (%)	1282 (12.11)	413 (10.73)	869 (12.90)	.001	0.17
Hispanic ethnicity, n (%)	2067 (19.52)	621 (16.13)	1446 (21.46)	<.001	1.17
Household income, ^a mean (SD)	7.54 (2.31)	7.86 (2.11)	7.35 (2.40)	<.001	8.03
Parental highest education, ^b n (%)				<.001	0.19
High school or below	449 (4.24)	135 (3.51)	314 (4.66)	NA	NA
High school graduate	976 (9.22)	229 (5.95)	747 (11.08)	NA	NA
Post-high school education	2571 (24.28)	713 (18.52)	1858 (27.57)	NA	NA
Bachelor's	2738 (25.86)	1060 (27.54)	1678 (24.90)	NA	NA
Master's or above	3834 (36.21)	1708 (44.38)	2126 (31.55)	NA	NA
Ownership of other devices					
Tablet, n (%)	4696 (44.35)	1896 (49.26)	2800 (41.55)	<.001	0
Laptop, n (%)	4026 (38.02)	1009 (26.21)	3017 (44.77)	<.001	0
iPod, n (%)	1144 (10.80)	719 (18.68)	425 (6.31)	<.001	0
Smartwatch, n (%)	619 (5.85)	109 (2.83)	510 (7.57)	<.001	0
Clinical characteristics at 2-y follow-up					
Psychopathology symptoms, mean (SD)	53.71 (5.64)	53.35 (5.46)	53.91 (5.73)	<.001	11.46
Clinical-threshold psychopathology, n (%)	906 (8.56)	289 (7.51)	617 (9.16)	<.001	11.46
Lifetime depression diagnosis, n (%)	604 (5.70)	174 (4.52)	430 (6.38)	<.001	0.71
BMI, kg/m ² , mean (SD)	20.75 (4.99)	19.84 (4.43)	21.30 (5.21)	<.001	16.10
Obesity, n (%)	1649 (15.57)	454 (11.80)	1195 (17.73)	<.001	16.13
Sleep duration, mean (SD), h	9.12 (1.26)	9.40 (1.15)	8.97 (1.29)	<.001	0.96
Insufficient sleep (<9h), n (%)	4372 (41.29)	1194 (31.02)	3178 (47.16)	<.001	0.96
Abbreviations: BMI, body mass index; NA, not applicable.					
Univariable comparisons were conducted for descriptive purposes to compare youths who did vs those who did not own a smartphone: Student <i>t</i> test for normally distributed continuous variables, Mann-Whitney U test for nonnormal continuous variables, and χ^2 test for categorical variables. If the expected frequencies in any cell of the contingency table were less than 5, we used Fisher's exact tests.					
^a Income is expressed as the mean of the values from 10 self-reported income brackets: (1) <\$5000; (2) \$5000 to 12 000; (3) \$12 000 to 16 000; (4) \$16 000 to 25 000; (5) \$25 000 to 35 000; (6) \$35 000 to 50 000; (7) \$50 000 to 75 000; (8) \$75 000 to 100 000; (9) \$100 000 to 200 000; and (10) >\$200 000.					
^b Parental education represents the highest level of education attained by either parent.					

2-year follow-up (mean age of 12.02 years), 77.2% (7602 of 9853) at the 3-year follow-up (mean age of 12.91 years), and 89.1% (4094 of 4593) at the 4-year follow-up (mean age of 14.08 years). The median (IQR) age of first smartphone acquisition was 11 (10, 11) years (range, 4–13; Figure 1).

Analysis 1: Smartphone Ownership and Health Outcomes at Age 12 Years

Compared with their peers who did not own smartphones, participants who owned smartphones tended to be slightly older, were at a more advanced pubertal stage, were more likely to be female, were more likely to identify as Black race

TABLE 2. Comparison of Youth Without Smartphones at 2-Year Follow-Up: Those Who Acquired a Smartphone by 3-Year Follow-Up vs Those Who Did Not

	Overall Sample	No Smartphone at Follow-Up	Owns a Smartphone at Follow-Up	Uncorrected <i>P</i> Value	Missing Data, %
<i>n</i>	3486	1940	1546	NA	NA
Smartphone type, iPhone	NA	NA	1017	NA	NA
Smartphone type, Android	NA	NA	529	NA	NA
Demographic characteristics					
Age at 2-y follow-up, mean (SD)	11.82 (0.63)	11.79 (0.62)	11.85 (0.64)	.006	0.03
Late/post-pubertal (2-y follow-up, <i>n</i> (%))	416 (11.93)	203 (10.46)	213 (13.78)	.004	4.96
Female sex assigned at birth, <i>n</i> (%)	1458 (41.82)	775 (39.95)	683 (44.18)	.01	0.03
White race, <i>n</i> (%)	2897 (83.10)	1640 (84.54)	1257 (81.31)	.02	0.2
Black race, <i>n</i> (%)	421 (12.08)	200 (10.31)	221 (14.29)	<.001	0.2
Asian, <i>n</i> (%)	252 (7.23)	164 (8.45)	88 (5.69)	.002	0.2
American Indian and Alaska Native race, <i>n</i> (%)	94 (2.70)	53 (2.73)	41 (2.65)	.98	0.2
Mixed race, <i>n</i> (%)	383 (10.99)	220 (11.34)	163 (10.54)	.51	0.2
Hispanic ethnicity, <i>n</i> (%)	552 (15.83)	276 (14.23)	276 (17.85)	.004	1.49
Household income, ^a mean (SD)	7.94 (2.09)	8.00 (2.00)	7.87 (2.19)	.09	6.63
Parental highest education, ^b <i>n</i> (%)				<.001	0.2
High school or below	111 (3.18)	52 (2.68)	59 (3.82)	NA	NA
High school graduate	187 (5.36)	87 (4.48)	100 (6.47)	NA	NA
Post high school education	614 (17.61)	299 (15.41)	315 (20.38)	NA	NA
Bachelor's	985 (28.26)	554 (28.56)	431 (27.88)	NA	NA
Master's or above	1582 (45.38)	941 (48.51)	641 (41.46)	NA	NA
Clinical characteristics at 2-y follow-up					
Psychopathology symptoms, mean (SD)	53.31 (5.46)	53.17 (5.38)	53.48 (5.57)	.11	11.42
Clinical-threshold psychopathology, <i>n</i> (%)	266 (7.63)	140 (7.22)	126 (8.15)	.44	11.42
Lifetime depression diagnosis, <i>n</i> (%)	150 (4.30)	86 (4.43)	64 (4.14)	.73	0.72
BMI, kg/m ² , mean (SD)	19.76 (4.34)	19.44 (4.07)	20.14 (4.62)	<.001	12.79
Obesity, <i>n</i> (%)	408 (11.70)	198 (10.21)	210 (13.58)	.005	12.82
Sleep duration, mean (SD), h	9.40 (1.14)	9.47 (1.12)	9.30 (1.15)	<.001	0.86

Abbreviations: BMI, body mass index; NA, not applicable.

Univariable comparisons were conducted for descriptive purposes: Student *t* test for normally distributed continuous variables, Mann-Whitney *U* test for nonnormal continuous variables, and χ^2 test for binary measures. If the expected frequencies in any cell of the contingency table were less than 5, we used Fisher's exact tests.

^a Income is expressed as the mean of the values from 10 self-reported income brackets: (1) <\$5000; (2) \$5000 to 12 000; (3) \$12 000 to 16 000; (4) \$16 000 to 25 000; (5) \$25 000 to 35 000; (6) \$35 000 to 50 000; (7) \$50 000 to 75 000; (8) \$75 000 to 100 000; (9) \$100 000 to 200 000; and (10) >\$200 000.

^b Parental education represents the highest level of education attained by either parent.

or of Hispanic ethnicity, and were more likely to be from households with lower income and lower parental educational attainment (Table 1). Participants with smartphones reported more depression, obesity, and insufficient sleep (Table 1). Correlations among psychopathology, BMI, and sleep were small (Pearson's *r* values ≤ 0.20 , all $P < .001$; Supplemental Figure S1).

In adjusted models (Figure 2A; Supplemental Tables S3–S5), owning a smartphone was associated with depression (OR 1.31, 95% CI: 1.05–1.63, $P = .016$), obesity (OR 1.40, 95% CI: 1.20–1.63, $P < .001$), and insufficient sleep (OR 1.62, 95% CI: 1.46–1.79, $P < .001$).

Analysis 2: Age of First Smartphone Acquisition and Health Outcomes at Age 12 Years

In adjusted models (Figure 2B; Supplemental Tables S6–S8), younger age of first smartphone acquisition was associated with obesity (for each earlier year of acquisition, OR 1.09, 95% CI: 1.02–1.16; $P = .007$) and insufficient sleep (OR 1.08, 95% CI: 1.02–1.12, $P = .005$), but not with depression (OR 1.03, 95% CI: 0.94–1.12; $P = .503$). Note that results from the base model, which only covaried for age and sex, showed an association of earlier age of smartphone ownership with depression, but this did not

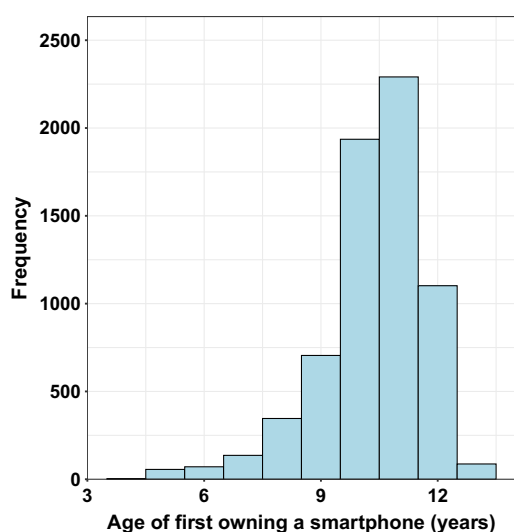


FIGURE 1. Distribution of age of smartphone acquisition. Histogram depicting the distribution of age at which ABCD Study participants first owned a smartphone as reported at the 2-year follow-up ABCD assessment wave. Abbreviation: ABCD, Adolescent Brain Cognitive Development.

satisfy the P value threshold (OR 1.08, 95% CI: 1.0–1.15; $P = .046$).

Analysis 3: Recent (Past-Year) Acquisition of a Smartphone and Health Outcomes at Age 13 Years

We analyzed data from 3486 youth who did not own smartphones at the 2-year follow-up and who also participated in the 3-year follow-up, when 1546 had acquired a smartphone and 1940 had not. Table 2 details comparison of

the groups. In adjusted models (Supplemental Tables S9–S11), youth who acquired a smartphone by the 3-year follow-up were more likely to reach clinical levels of psychopathology (OR 1.57, 95% CI: 1.12–2.20; $P = .009$; Figure 3) and to sleep less than 9 hours per night (OR 1.50, 95% CI: 1.26–1.77; $P < .001$; Figure 3), with no statistically significant difference in obesity (OR 1.57, 95% CI: 0.82–3.04; $P = .176$; Figure 3), compared with those who did not.

Sex Differences

We did not detect significant interactions with sex in any of the smartphone exposures.

Sensitivity Analyses

Analyses excluding participants who engage in problematic smartphone use yielded similar results to main analyses in direction and in statistical significance in all adjusted models (Supplemental Table S12). E-value analyses addressing unobserved confounding run on fully adjusted models suggested that implausibly strong unmeasured confounders would be required to explain away the observed associations (Supplemental Table S13). Analyses that used continuous outcomes (overall psychopathology BPM score, BMI, hours of sleep; Supplemental Tables S14–S22) and analyses that included longitudinal data from the 2-year, 3-year, and 4-year follow-up assessment waves (Supplemental Figure S2, Supplemental Tables S23–S28) revealed similar results in direction and statistical significance as main analyses, except for smartphone-by-sex interaction that became significant in association with

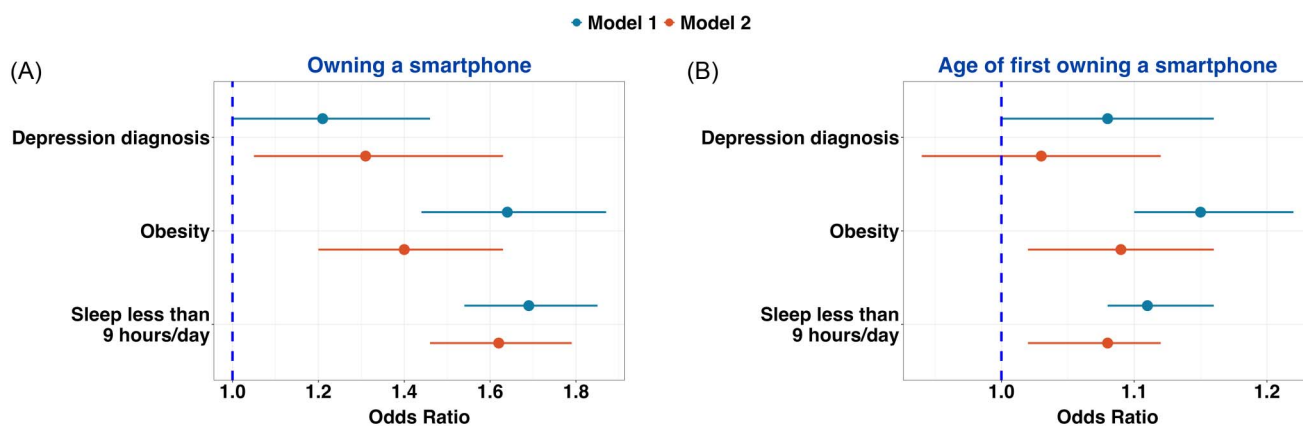


FIGURE 2. (A) Association of smartphone ownership and (B) age of first smartphone acquisition with depression, obesity, and insufficient sleep as captured at the 2-year follow-up assessment wave of ABCD Study (age 12 years). Model 1 adjusted for age and sex. Model 2 further adjusted for race, ethnicity, income, parent education, ownership of other devices (eg, tablet, laptop, iPod, smartwatch), pubertal status, and parental monitoring. Dots represent OR and 95% CI values. Vertical dashed lines represent the null hypothesis for no association between smartphone ownership and outcome. Note that, for visualization purposes, coefficients in panel B were transformed to reflect a positive relationship between younger age and adverse outcome (more depression, more obesity, insufficient sleep) by using the inverse of the OR and 95% CI values. Abbreviations: ABCD, Adolescent Brain Cognitive Development; OR, odds ratio.

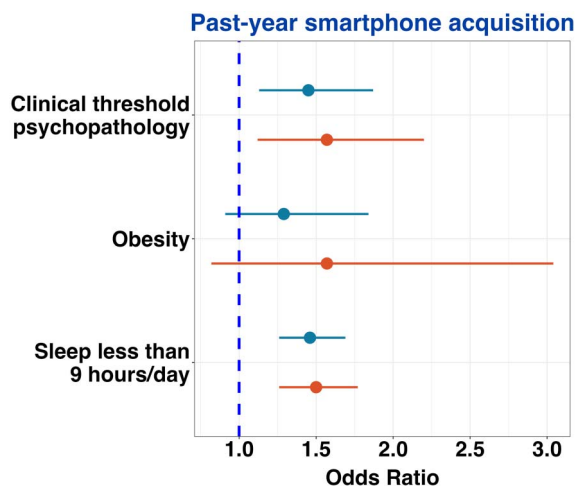


FIGURE 3.

Association of first smartphone acquisition over the past-year with clinical-threshold psychopathology, obesity, and insufficient sleep as captured at the 3-year follow-up assessment wave of ABCD Study (age 13 years). Model 1 adjusted for age and sex. Model 2 further adjusted for race, ethnicity, income, parent education, ownership of other devices (eg, tablet, laptop, iPod, smartwatch), pubertal status, parental monitoring, and the measurement of the index outcome a year earlier, ie, at the 2-year follow-up assessment. Psychopathology was measured with the self-reported BPM (age and sex standardized T-score), clinical threshold was considered for BPM score greater than or equal to 65. Dots represent OR and 95% CI values. Vertical dashed lines represent the null hypothesis for no association between past-year smartphone acquisition and outcome. Abbreviations: ABCD, Adolescent Brain Cognitive Development; BPM, brief problem monitor; OR, odds ratio.

sleep (Supplemental Table 25; interaction $P = .001$), with girls showing stronger associations between smartphone ownership and insufficient sleep than boys.

DISCUSSION

In this large-scale, multisite study of US youths of early adolescent age, we provide converging evidence for adverse associations of smartphone ownership with health outcomes, accounting for several demographic, socioeconomic, and parental factors. Smartphone ownership at age 12 years was related to depression, obesity, and insufficient sleep (Analysis 1). Earlier age of smartphone acquisition was associated with obesity and insufficient sleep at age 12 years (Analysis 2). Recent (past-year) first-time acquisition of smartphones by age 13 years was associated with clinical-threshold psychopathology and insufficient sleep, even when accounting for psychopathology and sleep in the prior year (Analysis 3). Results appear specific to smartphones, as our models accounted for ownership of other devices including tablets, laptops, iPods, and smartwatches. These findings contribute to a growing body of research indicating that an earlier age of smartphone ownership may be linked to poorer mental health outcomes. Prior studies have

reported that early smartphone acquisition is associated with greater psychological distress among Spanish teens³⁴ and that receiving a smartphone before age 13 years is associated with worse social and emotional health in young adulthood compared with those who received a smartphone after age 13 years in a multinational sample.³⁵ Notably, the only US study on this topic, to our knowledge, focused on Hispanic youths with low socioeconomic status with overweight or obesity and found no adverse association between smartphone ownership and either mental health or sleep.³⁶ This contrast highlights that findings from a narrow population may not apply broadly, strengthening the generalizability of our findings that were derived from a diverse US cohort from a multisite study.

No clear evidence exists to explain the deleterious associations between smartphone ownership and youth development. However, there is evidence that smartphone use contributes to fragmented attention, increased checking behaviors (which may impact relationships), and extended use (particularly in the evenings), which may lead to both mental and physical health challenges and less sleep,^{9,10} especially during a period of development during which many youths may not have sufficiently mature self-regulatory skills to make optimal choices regarding their smartphone use.³⁷ Importantly, our results are separable from problematic smartphone use, as our sensitivity analyses that excluded youths with problematic smartphone use suggest that smartphone ownership, even when not rising to the level of problematic use, may negatively impact adolescent mental and physical health, which has substantial public health implications.

A key finding is that an earlier age of first smartphone acquisition was associated with higher rates of obesity and insufficient sleep at age 12 years in models that adjusted for age, sex, and multiple confounders. Association of earlier age of smartphone ownership with higher rates of depression when adjusting for age and sex ($P = .046$) did not satisfy our P value threshold. We highlight the developmental context of these results. Prior literature that focused on addictive or excessive smartphone use has suggested that smartphone ownership around childhood or early adolescence may displace time spent on in-person interactions,³⁸ physical activity,³⁹ and restorative sleep,⁴⁰ key foundations for adolescent health. Notably, although the effect sizes for the age of first smartphone acquisition are small (with ORs for all 3 outcomes around 1.1 for each earlier year of ownership), adolescence is a sensitive period during which even modest subclinical mental health symptoms, overweight, and sleep problems can have long-term health consequences into adulthood.⁴¹⁻⁴³ Thus, small effect sizes at the individual level may translate to substantial public health significance. Another important developmental consideration involves sex differences during this age range, which coincide with the transition to puberty. Previous

research in older adolescents has suggested that girls may experience stronger adverse effects of smartphone use, particularly in relation to mental health outcomes such as depression.¹⁴ However, our exploratory analyses did not detect significant smartphone ownership by sex differences in association with depression in early adolescence. It is possible that such differences emerge later in adolescence, coinciding with the increase in depression rates among girls following the onset of puberty.⁴⁴ Note that our sensitivity analysis that leveraged data from multiple time points, increasing the number of observations, did reveal a significant smartphone-by-sex interaction in association with impaired sleep, suggesting greater susceptibility among girls. Future longitudinal work from early to late adolescence should reexamine sex differences in the ways that teens use their smartphones and how this relates to health outcomes.

Given our findings, it is evident that a concrete framework is needed to advise on childhood and early adolescent smartphone ownership to support the healthier development of youth. There are existing tools available to gauge youth readiness. For example, the AAP has developed an easily accessible questionnaire, ie, the PhoneReady Questionnaire,⁴⁵ which is a thoughtful instrument addressing a range of key youth (eg, need, judgement, impulsiveness, accountability) and parental factors (eg, readiness to impose boundaries) to consider prior to determining whether a child is ready to have a smartphone. Although the AAP and AACAP offer other *common-sense* guidelines for instituting safer practices, empirical research is needed to establish clearer guidelines specific to the following: (1) optimal age of ownership, (2) type of phone (eg, starter phone with more limited capabilities), and (3) uses (eg, limiting to texts, phone calls, restricting access to specific apps).

Study strengths include a large longitudinal cohort from diverse communities, but there are also notable limitations. The focus of this study was on smartphone ownership, but it did not provide insight into *what* youth were doing on their smartphones. Factors such as patterns of use and exposure to adverse content may contribute to health outcomes in ways that are distinct from age. Additionally, although our findings largely persisted when accounting for parental monitoring, there may be other important, unexplored factors that buffer against negative health outcomes. Because this study used observational data (a randomized placebo-controlled trial would not have been possible), we cannot be certain that unmeasured factors are not already more prevalent among youth without smartphones, prompting caution when drawing causal conclusions from our findings. We address the research question within these limitations via converging analyses, careful inclusion of key confounders, and sensitivity analysis for unobserved confounding.

Although our findings may signal a need for public policy changes, we are not advocating for restrictions on youth access and ownership of smartphones. Rather, we highlight the need for further research to better understand the adverse effects of smartphone ownership during the transition from childhood to adolescence, a developmental period that already presents a high risk for psychiatric disorders^{46,47} and sleep problems.^{43,48} More broadly, smartphone ownership offers unique challenges because it may grant youths unfettered access to a world for which they may not be ready, without the discipline to effectively manage their own use. Like youths' diet, media consumption (eg, television, social media), and relationships (eg, with peers, adults), smartphone ownership requires dedicated oversight. This is particularly true because there may be various adverse consequences and challenges for certain youths who *do not* own smartphones, which underscores the need to protect these youths who require smartphones and support families as they embark on this rite of passage.

ACKNOWLEDGMENTS

Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive Development (ABCD) Study (<https://abcdstudy.org>), held in the National Institute for Mental Health Data Archive. This is a multisite, longitudinal study designed to recruit more than 10 000 children age 9 to 10 years and follow them over 10 years into early adulthood. The ABCD Study is supported by the National Institutes of Health and additional federal partners under award numbers U01DA041048, U01DA050989, U01DA051016, U01DA041022, U01DA051018, U01DA051037, U01DA050987, U01DA041174, U01DA041106, U01DA041117, U01DA041028, U01DA041134, U01DA050988, U01DA051039, U01DA041156, U01DA041025, U01DA041120, U01DA051038, U01DA041148, U01DA041093, U01DA041089, U24DA041123, and U24DA041147. A full list of supporters is available at the following link: <https://abcdstudy.org/federal-partners.html>. A listing of participating sites and a complete listing of the study investigators can be found at the following link: https://abcdstudy.org/consortium_members/. ABCD Study consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in the analysis or writing of this report. This manuscript reflects the views of the authors and may not reflect the opinions or views of the National Institutes of Health or ABCD Study consortium investigators. The authors thank Christina Ly for supporting the literature review for this project.

ABBREVIATIONS

AAP: American Academy of Pediatrics
AACAP: American Academy of Child and Adolescent Psychiatry
ABCD: Adolescent Brain Cognitive Development
BMI: body mass index
BPM: Brief Problem Monitor

IRB: institutional review board
KSADS: Kiddie Schedule for Affective Disorders and Schizophrenia Child Psychiatric Diagnostic Interview

OR: odds ratio
STROBE: Strengthening the Reporting of Observational Studies in Epidemiology

Dr Auerbach contributed to study conceptualization and writing the first draft of the manuscript. All authors substantially contributed to and approved the final version of the manuscript. Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive Development Study (<https://abcdstudy.org>), held in the National Institute of Mental Health Data Archive and available for researchers on application (<https://nda.nih.gov/abcd/request-access>). Analytical code describing all analyses is available at the following link: https://github.com/barzilab1/ABCD_smartphone_health.

CONFLICT OF INTEREST DISCLOSURES: Dr Barzilay serves on the Advisory Board and reports stock ownership in Taliaz Health, with no relevance to this work. Ms Visoki's spouse is a shareholder and executive in Kidas, with no conflict of interest relevant to this work. In the past 3 years, Dr Auerbach has received consulting fees and equity from Get Sonar, Inc. He has also received consulting fees from RPA Health Consulting, Inc., and Covington & Burling LLP, which is representing a social media company in litigation. He has received funding from the Morgan Stanley Foundation, the National Institute of Mental Health, and the Tommy Fuss Fund. All other authors report no biomedical financial interests or potential conflicts of interest. There are no conflicts of interest with the work conducted in this study. All views expressed are solely those of the authors.

FUNDING: Dr Barzilay was supported by R01MH134886 and P50MH115838 from the National Institute of Mental Health and by SRG-006-22 from the American Foundation for Suicide Prevention. Dr Auerbach was partially supported by R01MH135488. The funder had no role in study design, data analysis, manuscript preparation, or the decision to submit the manuscript for publication.

Accepted for Publication Date: October 13, 2025

<https://doi.org/10.1542/peds.2025-072941>

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REFERENCES

1. Rideout V. *Common Sense Census: Media Use by Tweens and Teens, 2019*. Common Sense Media; 2019. Accessed November 5, 2024. <https://www.commonsensemedia.org/research/the-common-sense-census-media-use-by-tweens-and-teens-2019>
2. Rideout V, Peebles A, Mann S, Robb MB. *Common Sense Census: Media Use by Tweens and Teens, 2021*. Common Sense Media; 2022.
3. Boers E, Afzali MH, Newton N, Conrod P. Association of screen time and depression in adolescence. *JAMA Pediatr*. 2019;173(9):853–859. PubMed doi: 10.1001/jamapediatrics.2019.1759
4. Hardy LL, Denney-Wilson E, Thrift AP, Okely AD, Baur LA. Screen time and metabolic risk factors among adolescents. *Arch Pediatr Adolesc Med*. 2010;164(7):643–649. PubMed doi: 10.1001/archpediatrics.2010.88
5. American Academy of Child and Adolescent Psychiatry. Screen time and children. Facts for families guide. [aacap.org](https://www.aacap.org/AACAP/Families_and_Youth/Facts_for_Families/FFF-Guide/Children-And-Watching-TV-054.aspx). May 2024. Accessed November 5, 2024. https://www.aacap.org/AACAP/Families_and_Youth/Facts_for_Families/FFF-Guide/Children-And-Watching-TV-054.aspx
6. Richter A, Adkins V, Selkie E. Youth perspectives on the recommended age of mobile phone adoption: survey study. *JMIR Pediatr Parent*. 2022;5(4):e40704. PubMed doi: 10.2196/40704
7. Ko Y, Park S. Impacts of problematic smartphone use on children: perspectives from main caregivers. *Arch Psychiatr Nurs*. 2023;46:59–64. PubMed doi: 10.1016/j.apnu.2023.08.007
8. Kliesener T, Meigen C, Kiess W, Poulain T. Associations between problematic smartphone use and behavioural difficulties, quality of life, and school performance among children and adolescents. *BMC Psychiatry*. 2022;22(1):195. PubMed doi: 10.1186/s12888-022-03815-4
9. Kim MH, Min S, Ahn JS, An C, Lee J. Association between high adolescent smartphone use and academic impairment, conflicts with family members or friends, and suicide attempts. *PLoS One*. 2019;14(7):e0219831. PubMed doi: 10.1371/journal.pone.0219831
10. Fischer-Grote L, Kothgassner OD, Felnhöfer A. Risk factors for problematic smartphone use in children and adolescents: a review of existing literature. *Neuropsychiatr*. 2019;33(4):179–190. PubMed doi: 10.1007/s40211-019-00319-8
11. Pagliaccio D, Tran KT, Visoki E, DiDomenico GE, Auerbach RP, Barzilay R. Probing the digital exposome: associations of social media use patterns with youth mental health. *NPP Digit Psychiatry Neurosci*. 2024;2(1):1–10. doi: 10.1038/s44277-024-00006-9
12. Xiao Y, Meng Y, Brown TT, Keyes KM, Mann JJ. Addictive screen use trajectories and suicidal behaviors, suicidal ideation, and mental health in US youths. *JAMA*. 2025;334(3):219–228. PubMed doi: 10.1001/jama.2025.7829
13. Sohn S, Rees P, Wildridge B, Kalk NJ, Carter B. Prevalence of problematic smartphone usage and associated mental health outcomes amongst children and young people: a systematic review, meta-analysis and GRADE of the evidence. *BMC Psychiatry*. 2019;19(1):1–10. PubMed doi: 10.1186/S12888-019-2350-X/FIGURES/3

14. Twenge JM, Martin GN. Gender differences in associations between digital media use and psychological well-being: evidence from three large datasets. *J Adolesc*. 2020;79(1):91–102. PubMed doi: 10.1016/j.adolescence.2019.12.018
15. Fischer-Grote L, Kothgassner OD, Felnhofer A. The impact of problematic smartphone use on children's and adolescents' quality of life: a systematic review. *Acta Paediatr*. 2021;110(5):1417–1424. PubMed doi: 10.1111/apa.15714
16. Karcher NR, Barch DM. The ABCD study: understanding the development of risk for mental and physical health outcomes. *Neuropsychopharmacology*. 2020;46(1):131–142. doi: 10.1038/s41386-020-0736-6
17. Townsend L, Kobak K, Kearney C, et al. Development of three web-based computerized versions of the Kiddie Schedule for Affective Disorders and Schizophrenia Child Psychiatric Diagnostic Interview: preliminary validity data. *J Am Acad Child Adolesc Psychiatry*. 2020;59(2):309–325. PubMed doi: 10.1016/j.jaac.2019.05.009
18. Piper BJ, Gray HM, Raber J, Birkett MA. Reliability and validity of brief problem monitor, an abbreviated form of the child behavior checklist. *Psychiatry Clin Neurosci*. 2014;68(10):759–767. PubMed doi: 10.1111/pcn.12188
19. Achenbach TM, McConaughy SH, Ivanova MY, Rescorla LA. *Manual for the ASEBA Brief Problem Monitor (BPM)*. Rescorla Research Center for Children, Youth, and Families University of Vermont; 2011.
20. Ogden CL, Flegal KM. Changes in terminology for childhood overweight and obesity. *Natl Health Stat Rep*. 2010;(25):1–5. PubMed
21. Paruthi S, Brooks LJ, D'Ambrosio C, et al. Consensus Statement of the American Academy of Sleep Medicine on the Recommended Amount of Sleep for Healthy Children: Methodology and Discussion. *J Clin Sleep Med*. 2016;12(11):1549–1561. PubMed doi: 10.5664/jcsm.6288
22. Argabright ST, Moore TM, Visoki E, DiDomenico GE, Taylor JH, Barzilay R. Association between racial/ethnic discrimination and pubertal development in early adolescence. *Psychoneuroendocrinology*. 2022;140:105727. PubMed doi: 10.1016/j.psyneuen.2022.105727
23. Zucker RA, Gonzalez R, Feldstein Ewing SW, et al. Assessment of culture and environment in the Adolescent Brain and Cognitive Development Study: rationale, description of measures, and early data. *Dev Cogn Neurosci*. 2018;32:107–120. PubMed doi: 10.1016/j.dcn.2018.03.004
24. Pelham WE III, Tapert SF, Gonzalez MR, et al. Parental knowledge/monitoring and depressive symptoms during adolescence: protective factor or spurious association? *Res Child Adolesc Psychopathol*. 2022;50(7):919–931. PubMed doi: 10.1007/s10802-021-00896-9
25. Kim KW, Wallander JL, Felt JM, Elliott MN, Schuster MA. Associations of parental general monitoring with adolescent weight-related behaviors and weight status. *Obesity (Silver Spring)*. 2019;27(2):280–287. PubMed doi: 10.1002/oby.22372
26. Gunn HE, O'Rourke F, Dahl RE, et al. Young adolescent sleep is associated with parental monitoring. *Sleep Health*. 2019;5(1):58–63. PubMed doi: 10.1016/j.sleh.2018.09.001
27. Lett E, Asabor E, Beltrán S, Cannon AM, Arah OA. Conceptualizing, contextualizing, and operationalizing race in quantitative health sciences research. *Ann Fam Med*. 2022;20(2):157–163. PubMed doi: 10.1370/afm.2792
28. Cuevas AG, Krobath DM, Rhodes-Bratton B, et al. Association of racial discrimination with adiposity in children and adolescents. *JAMA Netw Open*. 2023;6(7):e2322839–e2322839. PubMed doi: 10.1001/jamanetworkopen.2023.22839
29. Wang Y, Zhao Z, Zhang Y, et al. Race, ethnicity, and sleep in US children. *JAMA Netw Open*. 2024;7(12):e2449861–e2449861. PubMed doi: 10.1001/jamanetworkopen.2024.49861
30. Harris B, Regan T, Schueler J, Fields SA. Problematic mobile phone and smartphone use scales: a systematic review. *Front Psychol*. 2020;11:672. PubMed doi: 10.3389/fpsyg.2020.00672
31. Walsh SP, White KM, Young RM. Needing to connect: the effect of self and others on young people's involvement with their mobile phones. *Aust J Psychol*. 2010;62(4):194–203. doi: 10.1080/00049530903567229
32. Chu J, Ganson KT, Testa A, et al. Screen time, problematic screen use, and eating disorder symptoms among early adolescents: findings from the Adolescent Brain Cognitive Development (ABCD) Study. *Eat Weight Disord*. 2024;29(1):57. PubMed doi: 10.1007/s40519-024-01685-1
33. VanderWeele TJ, Ding P. Sensitivity analysis in observational research: Introducing the E-Value. *Ann Intern Med*. 2017;167(4):268–274. PubMed doi: 10.7326/M16-2607
34. Martín-Cárdaba MÁ, Martínez Díaz MV, Lafuente Pérez P, García Castro J. Smartphone ownership, minors' well-being, and parental mediation strategies. An analysis in the context of social media influencers. *J Youth Adolesc*. 2024;53(10):2202–2218. PubMed doi: 10.1007/s10964-024-02013-7
35. Thiagarajan TC, Newson JJ, Swaminathan S. Protecting the developing mind in a digital age: a global policy imperative. *J Human Dev Capabil*. 2025;26(3):493–504. doi: 10.1080/19452829.2025.2518313
36. Sun X, Haydel KF, Matheson D, Desai M, Robinson TN. Are mobile phone ownership and age of acquisition associated with child adjustment? A 5-year prospective study among low-income Latinx children. *Child Dev*. 2023;94(1):303–314. PubMed doi: 10.1111/cdev.13851
37. Abi-Jaoude E, Naylor KT, Pignatiello A. Smartphones, social media use and youth mental health. *CMAJ*. 2020;192(6):E136–E141. PubMed doi: 10.1503/cmaj.190434
38. Ihm J. Social implications of children's smartphone addiction: the role of support networks and social engagement. *J Behav Addict*. 2018;7(2):473–481. PubMed doi: 10.1556/2006.7.2018.48
39. Al-Amri A, Abdulaziz S, Bashir S, Ahsan M, Abualait T. Effects of smartphone addiction on cognitive function and physical activity in middle-school children: a cross-sectional study. *Front Psychol*. 2023;14:1182749. PubMed doi: 10.3389/fpsyg.2023.1182749

40. Kim SY, Han S, Park EJ, et al. The relationship between smartphone overuse and sleep in younger children: a prospective cohort study. *J Clin Sleep Med*. 2020;16(7):1133–1139. PubMed doi: 10.5664/jcsm.8446
41. Balázs J, Miklósi M, Keresztény A, et al. Adolescent subthreshold-depression and anxiety: psychopathology, functional impairment and increased suicide risk. *J Child Psychol Psychiatry*. 2013;54(6): 670–677. PubMed doi: 10.1111/jcpp.12016
42. Koskinen J, Magnussen CG, Sabin MA, et al. Youth overweight and metabolic disturbances in predicting carotid intima-media thickness, type 2 diabetes, and metabolic syndrome in adulthood: the Cardiovascular Risk in Young Finns study. *Diabetes Care*. 2014; 37(7):1870–1877. PubMed doi: 10.2337/dc14-0008
43. Kansagra S. Sleep disorders in adolescents. *Pediatrics*. 2020; 145(suppl 2):S204–S209. PubMed doi: 10.1542/peds.2019-2056l
44. Nolen-Hoeksema S. Gender differences in depression. *Curr Dir Psychol Sci*. 2001;10(5):173–176. doi: 10.1111/1467-8721.00142
45. AAP and AT&T launch free digital tools to help families. Telecom Ramblings Newswire. July 20, 2022. Accessed November 17, 2024. <https://newswire.telecomramblings.com/2022/07/aap-and-att-launch-free-digital-tools-to-help-families/>
46. Merikangas KR, He JP, Burstein M, et al. Lifetime prevalence of mental disorders in U.S. adolescents: results from the National Comorbidity Survey Replication–Adolescent Supplement (NCS-A). *J Am Acad Child Adolesc Psychiatry*. 2010;49(10):980–989. PubMed doi: 10.1016/j.jaac.2010.05.017
47. Solmi M, Radua J, Olivola M, et al. Age at onset of mental disorders worldwide: large-scale meta-analysis of 192 epidemiological studies. *Molecular Psychiatry* 2021 27:1. 2021;27(1):281–295. doi: 10.1038/s41380-021-01161-7
48. Owens J; Adolescent Sleep Working Group; Committee on Adolescence. Insufficient sleep in adolescents and young adults: an update on causes and consequences. *Pediatrics*. 2014;134(3): e921–e932. PubMed doi: 10.1542/peds.2014-1696